

Experiment No.	1
Experiment Title	Working with Python Packages – NumPy, SciPy, Scikit-Learn, Matplotlib ¹
Student Name	Sadhana J
Register Number	3122247001052
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	28.07.2026
Github Repository	Click Here

1 Objective

To explore core Python data-science libraries (NumPy, Pandas, SciPy, Scikit-learn, Matplotlib) and apply a complete machine learning workflow – loading, EDA, preprocessing, feature selection, splitting, model training and evaluation – on four benchmark datasets (Iris, Loan Amount Prediction, Pima Diabetes, Spambase) to identify the correct ML task type and algorithm for each.

2 Problem Statement

Given four heterogeneous public datasets, the task is to (i) identify whether each corresponds to a classification or regression problem, (ii) apply an appropriate feature-selection technique, (iii) build a baseline Random Forest model for each, and (iv) evaluate model performance on validation and test splits using standard metrics.

3 Dataset and Task Summary

Dataset	ML Task	Feature Selection	Suitable ML Algorithm
Iris Dataset	Classification (multi-class, 3 species)	Chi-Square (SelectKBest)	Random Forest Classifier
Loan Amount Prediction	Regression (continuous target)	ANOVA / f_regression	Random Forest Regressor
Predicting Diabetes	Classification (binary)	Chi-Square (SelectKBest)	Random Forest Classifier
Email Spam Classification	Classification (binary)	Chi-Square (SelectKBest)	Random Forest Classifier

4 Data Preprocessing (Common Pipeline)

Applied identically to all four datasets via a reusable `preprocessData()` function:

- Column names stripped; irrelevant identifier columns (e.g. `Loan_ID`) dropped.

- Rows with a missing target dropped; empty/constant columns removed; duplicate rows dropped.
- Missing numeric values imputed with median; missing categorical values imputed with mode.
- Categorical variables label-encoded; target label-encoded where categorical.
- Numeric features standardized using `StandardScaler`.
- Data split 70% train / 10% validation / 20% test using `train_test_split`.

5 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook (JupyterLab)
Operating System	
Hardware	
Model Used	RandomForestClassifier / RandomForestRegressor (random_state=42)

6 Part A: Iris Dataset (Classification)

Dataset Description

Dataset Name	Iris.csv
Number of Samples	150
Number of Features	6 (Id, SepalLengthCm, SepalWidthCm, PetalLengthCm, PetalWidthCm, Species)
Number of Classes	3 (Species)
Missing Values	None
Feature Selection	Chi-Square, k=4 – ranking: PetalWidthCm (96.46) > PetalLengthCm (90.01) > Id (77.50) > SepalLengthCm (49.63) > SepalWidthCm (24.10)
Train / Val / Test Split	104 / 15 / 31

EDA Figures

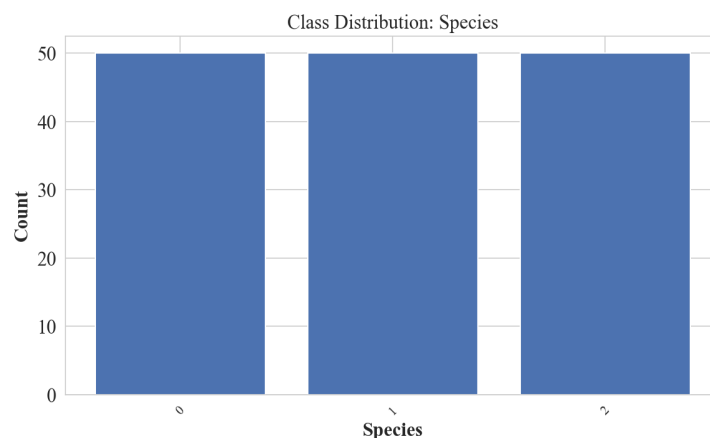


Figure 1: Class Distribution: Species

Inference: All three Iris species (Setosa, Versicolor, Virginica) are represented in roughly equal counts, so the dataset is balanced and does not require class-imbalance handling such as resampling or class weighting.

EDA Figures

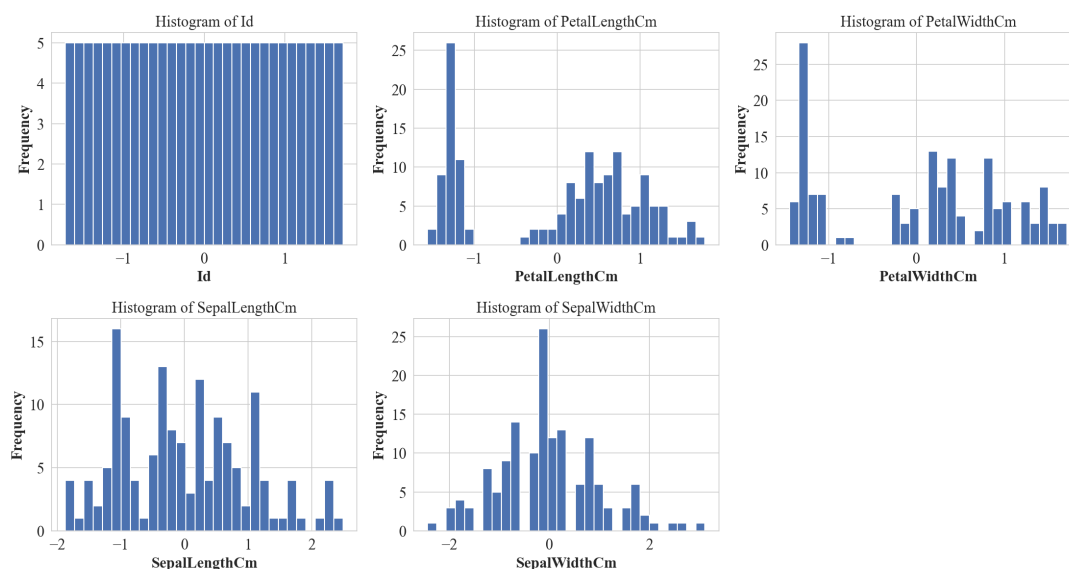


Figure 2: Histogram grid of the top features (Id, SepalLengthCm, SepalWidthCm, PetalLengthCm, PetalWidthCm)

Inference: Petal-related features show a clearly multi-modal distribution corresponding to the three species, whereas sepal width is closer to a single-peaked, roughly normal distribution.

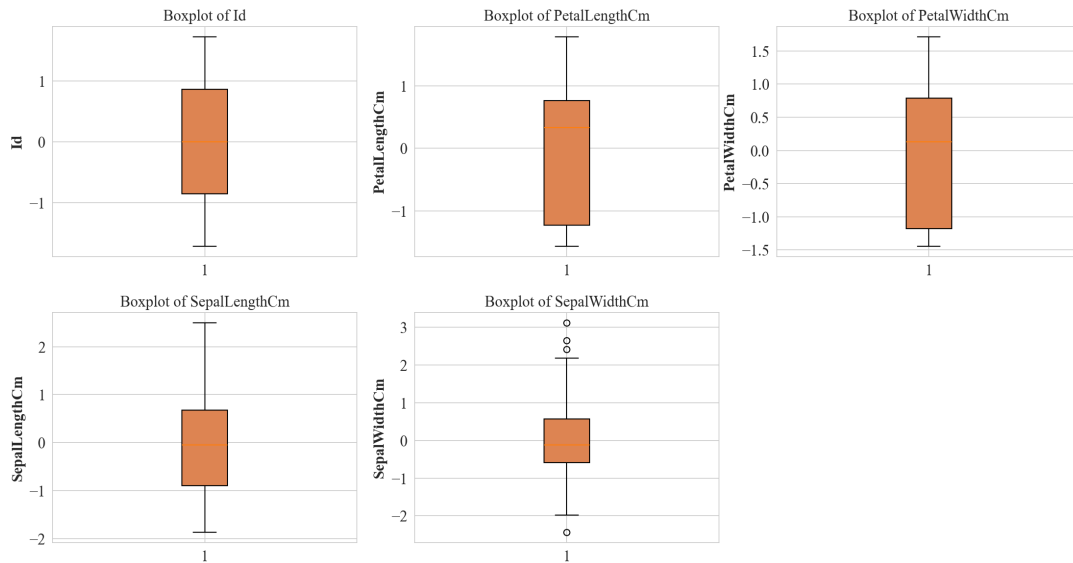


Figure 3: Boxplot grid of the top features

Inference: SepalWidthCm shows a small number of outliers beyond the whiskers, while the petal features have tighter, cleanly separated spreads, again suggesting strong class separability.

Results

Metric	Validation	Test
Accuracy	1.0000	1.0000
Precision (weighted)	1.0000	1.0000
Recall (weighted)	1.0000	1.0000
F1-score (weighted)	1.0000	1.0000

Inference: The Random Forest classifier achieved perfect accuracy on both validation (15/15) and test (31/31) splits, confirming that petal length/width alone make the three Iris species linearly separable.

7 Part B: Loan Amount Prediction (Regression)

Dataset Description

Dataset Name	train.csv (Loan Amount Prediction)
Number of Samples	30000 raw / 29660 after cleaning
Number of Features	24 (23 predictors + target), Customer_ID/Loan_ID dropped
Target Variable	Loan Sanction Amount (USD) – continuous
Missing Values	None after preprocessing (imputed)

Feature Selection	ANOVA f_regression, k=8 – top: Loan Amount Request (33132.6), Property Price (26535.6), Current Loan Expenses (9053.9), Credit Score (4407.6)
Train / Val / Test Split	20761 / 2966 / 5933

EDA Figures

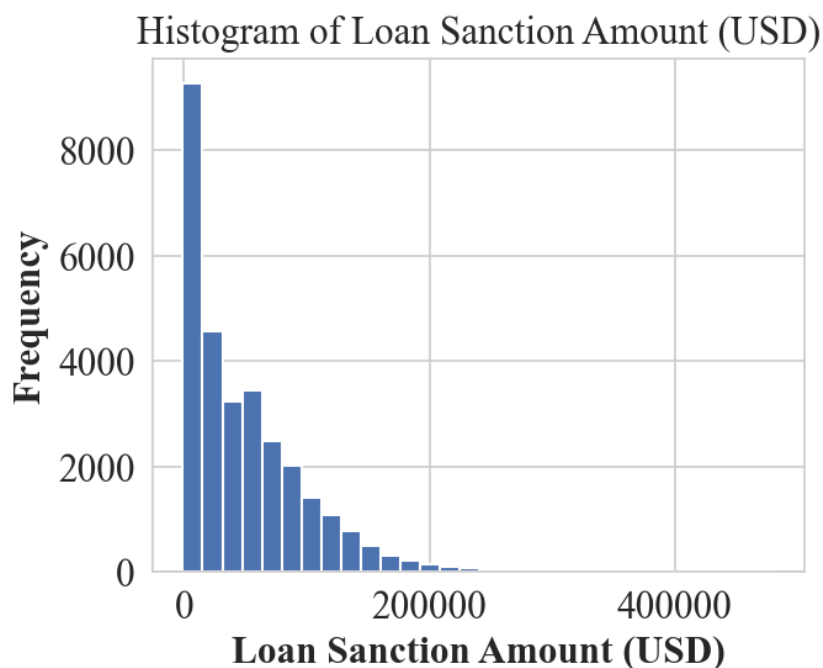


Figure 4: Histogram of Loan Sanction Amount (USD) – target distribution

Inference: The target variable is right-skewed with a large concentration of low-to-moderate sanctioned amounts and a long tail of high-value loans, motivating the use of standardized regression models.

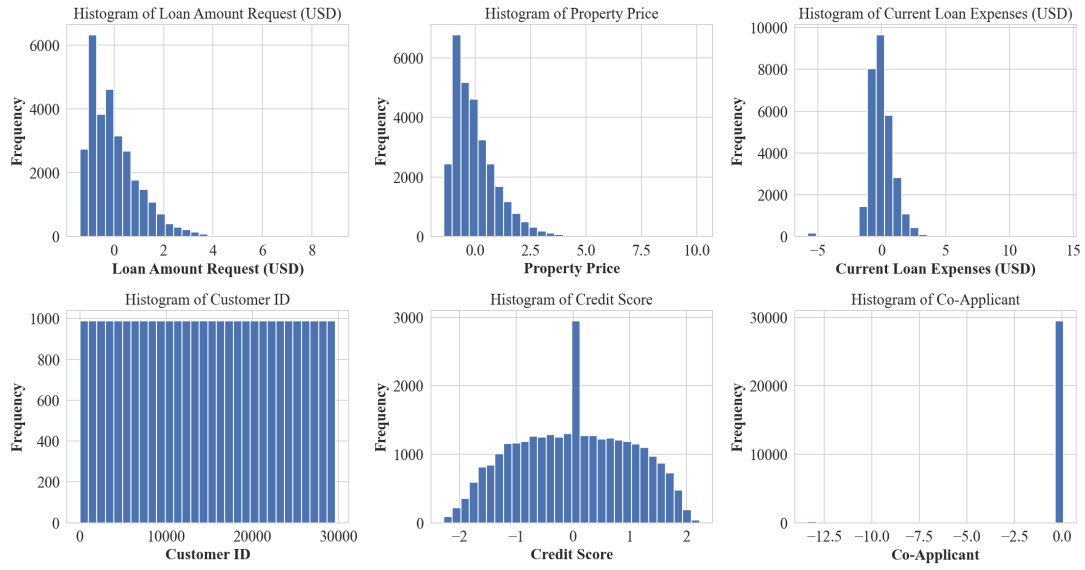


Figure 5: Histogram grid of the top 6 selected features

Inference: Income and loan-request amounts are heavily right-skewed, suggesting log-scaling could further stabilize variance, while Credit Score is closer to normally distributed.

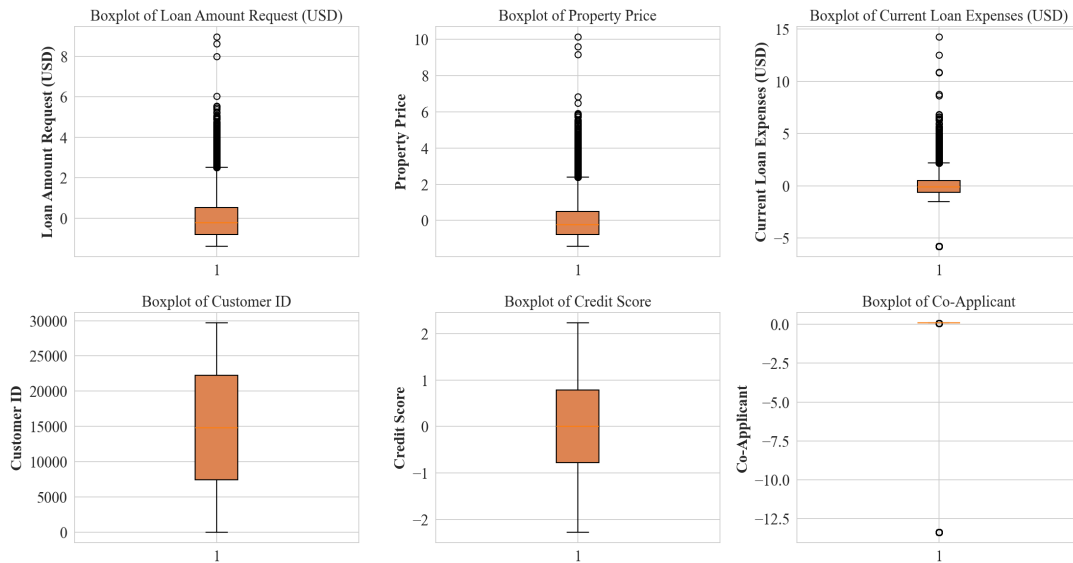


Figure 6: Boxplot grid of the top 6 selected features

Inference: Loan Amount Request and Property Price show a considerable number of high-value outliers, which likely contribute to the higher error on the test set relative to validation.

Results

Metric	Validation	Test
MAE	5591.36	5964.61
MSE	197804677.24	254135872.98
RMSE	14064.31	15941.64
R^2	0.9070	0.8918

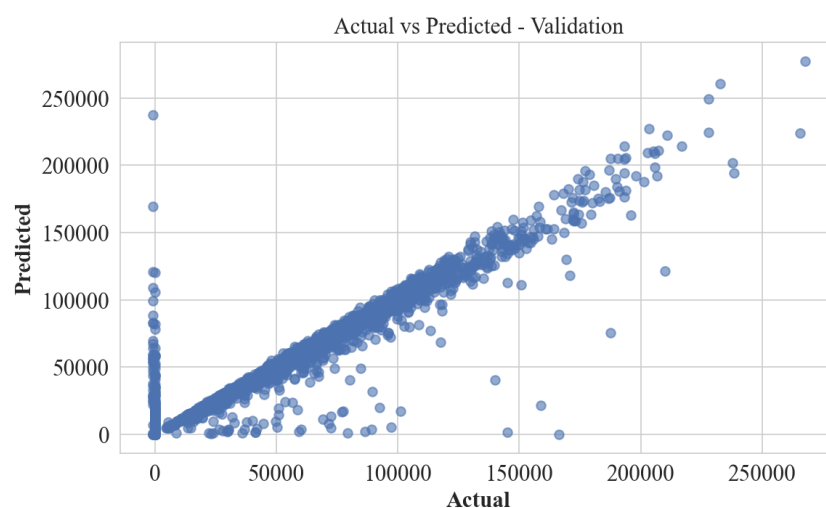


Figure 7: Actual vs Predicted – Validation

Inference: Predicted values track actual sanctioned amounts closely along the diagonal for the validation set, confirming the model captures the dominant loan-amount trend well ($R^2=0.907$).

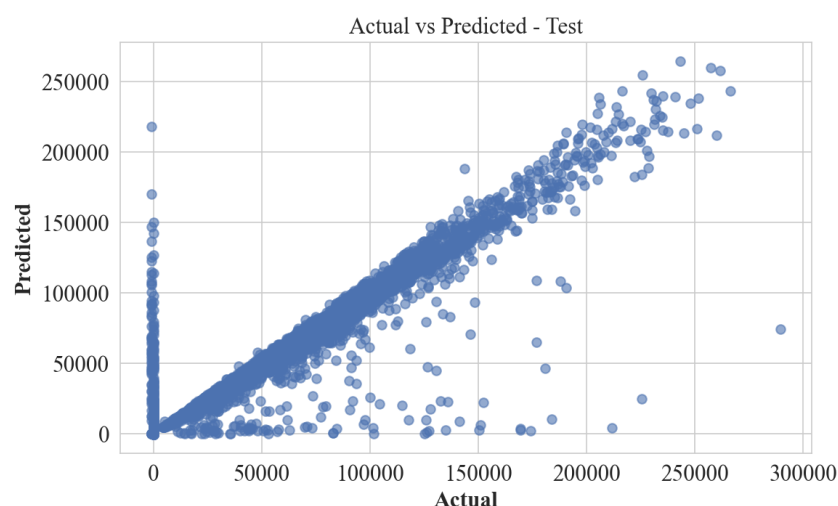


Figure 8: Actual vs Predicted – Test

Inference: The test scatter is slightly more dispersed than validation, particularly at higher loan amounts, indicating mild underprediction for large sanctioned loans – a common regression-to-the-mean effect with tree ensembles.

8 Part C: Predicting Diabetes (Classification)

Dataset Description

Dataset Name	diabetes.csv (Pima Indians Diabetes)
Number of Samples	768
Number of Features	9 (8 predictors + Outcome)
Number of Classes	2 (Outcome: 0/1)
Missing Values	None (zero-value entries represent missing physiological measurements, not imputed separately)
Feature Selection	Chi-Square, k=6 – top: Glucose (44.19), Age (41.89), Pregnancies (33.12), DiabetesPedigreeFunction (19.51)
Train / Val / Test Split	537 / 77 / 154

EDA Figures

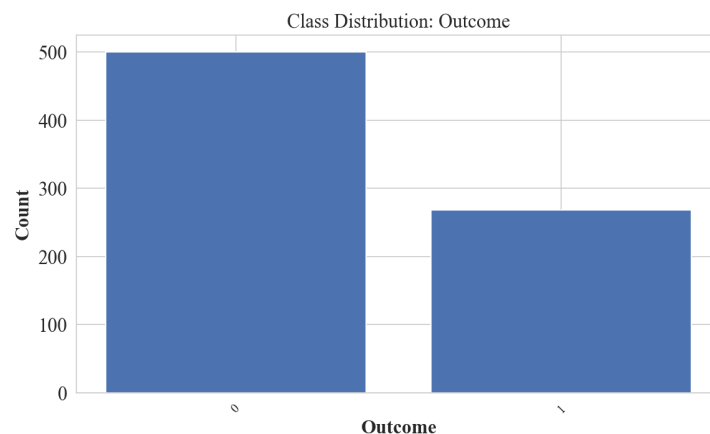


Figure 9: Class Distribution: Outcome

Inference: The dataset is imbalanced – roughly 65% non-diabetic vs 35% diabetic cases – which explains the lower recall/precision on the minority (diabetic) class compared to the majority class.

EDA Figures

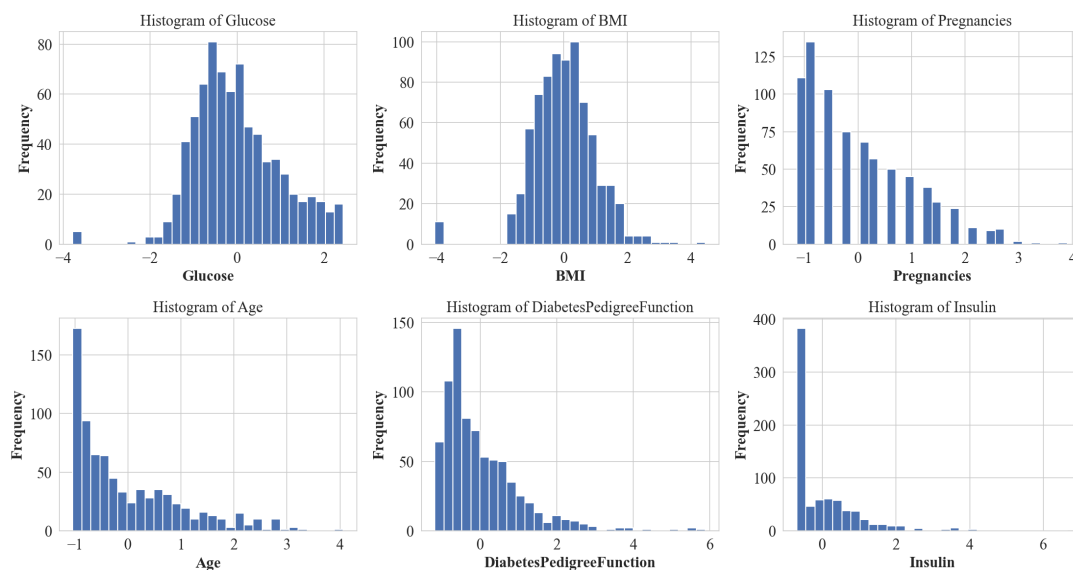


Figure 10: Histogram grid of the top 6 selected features

Inference: Glucose and BMI approximate a normal distribution, while Insulin and DiabetesPedigreeFunction are strongly right-skewed with several extreme values.

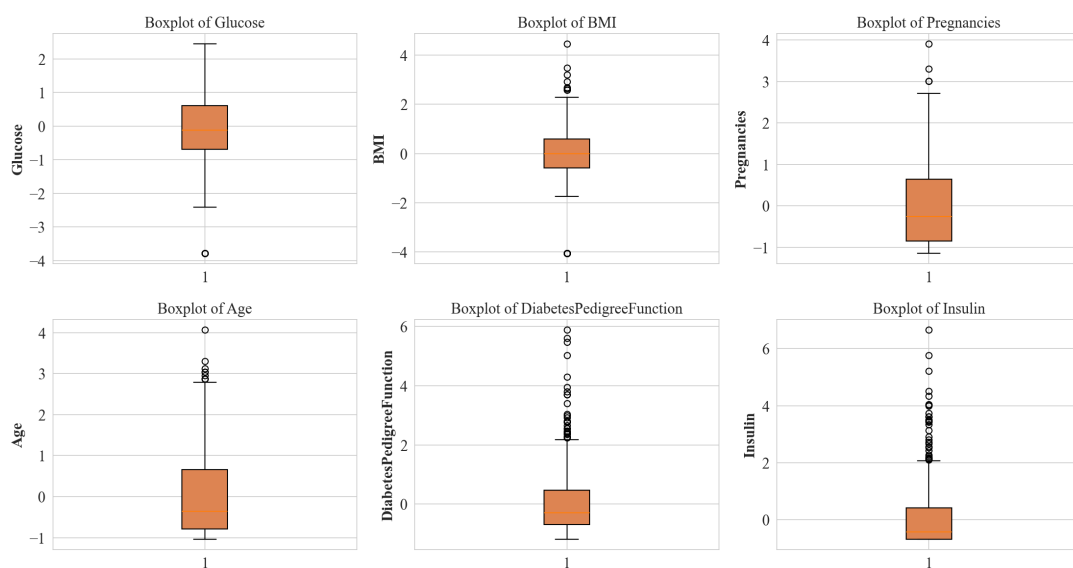


Figure 11: Boxplot grid of the top 6 selected features

Inference: Insulin and SkinThickness contain the most pronounced outliers, likely due to recorded zero values representing missing measurements rather than true zeros.

Results

Metric	Validation	Test
Accuracy	0.7403	0.7597
Precision (weighted)	0.7531	0.7608
Recall (weighted)	0.7403	0.7597
F1-score (weighted)	0.7447	0.7602

Inference: The model correctly identifies most non-diabetic cases but misclassifies a notable fraction of diabetic cases as negative (recall ≈ 0.67 on the diabetic class in the test set), reflecting the class imbalance and physiological overlap between classes – acceptable as a baseline but insufficient for clinical deployment without further tuning or resampling.

9 Part D: Email Spam Classification (Spambase)

Dataset Description

Dataset Name	spambase.csv
Number of Samples	4601 raw / 4210 after cleaning
Number of Features	58 (57 predictors + spam label)
Number of Classes	2 (spam / not spam)
Missing Values	None
Feature Selection	Chi-Square, k=10 – top: word_freq_remove (1592.4), word_freq_000 (1559.3), char_freq_\$ (1415.2), word_freq_free (1031.5)
Train / Val / Test Split	2946 / 421 / 843

EDA Figures



Figure 12: Class Distribution: spam

Inference: The dataset has a moderate class imbalance, with non-spam emails outnumbering spam emails, which is factored into the weighted precision/recall/F1 metrics used.

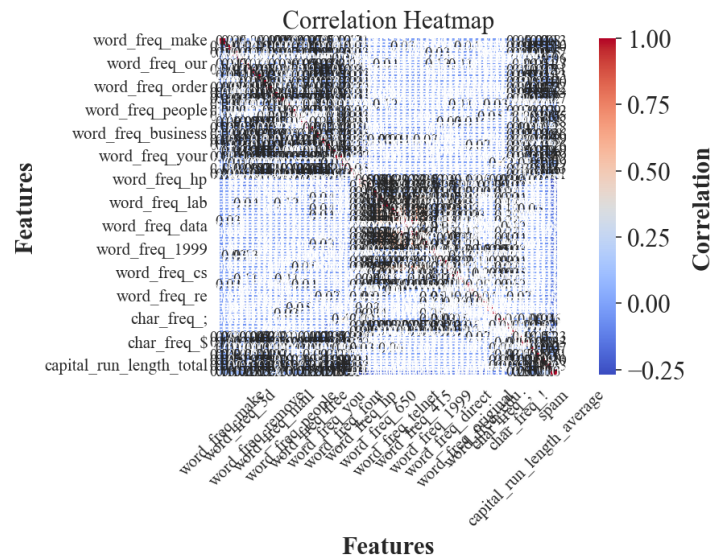


Figure 13: Correlation Heatmap

Inference: Word-frequency features such as "remove", "free", and "\$" show the strongest positive correlation with the spam label, matching common spam vocabulary patterns.

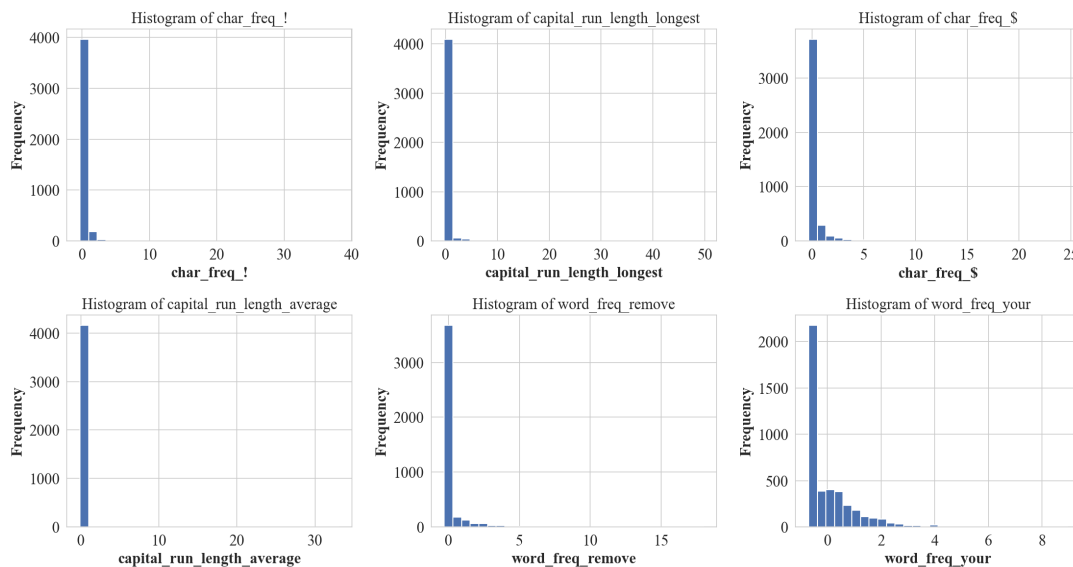


Figure 14: Histogram grid of the top 6 selected features

Inference: Nearly all word-frequency features are extremely right-skewed (mostly zero with rare high spikes), typical of sparse bag-of-words style features.

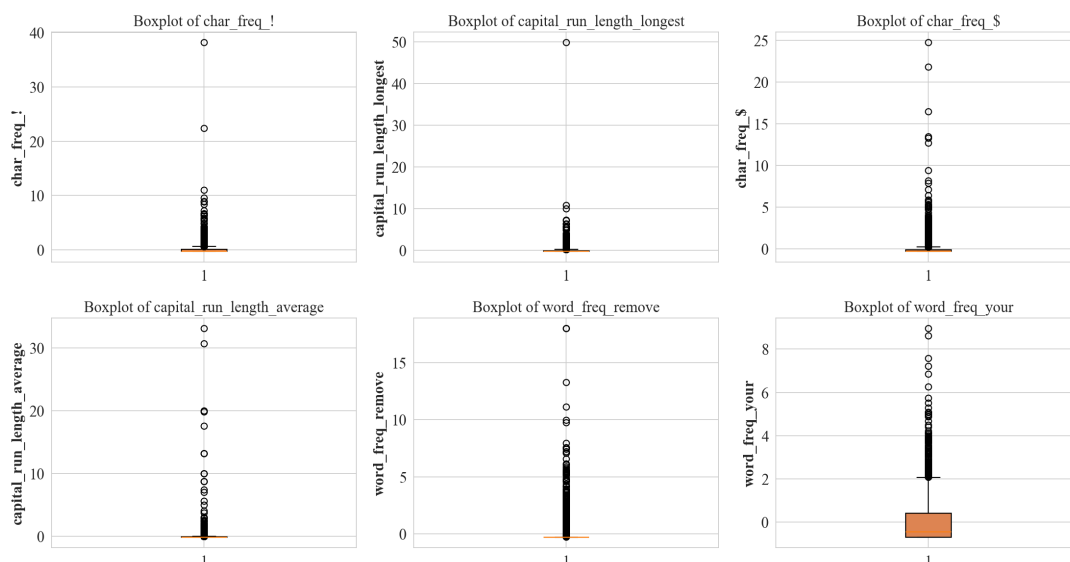


Figure 15: Boxplot grid of the top 6 selected features

Inference: Every selected feature shows a dense cluster near zero with a long tail of outliers, confirming that a handful of emails use these spam-indicative words very heavily.

Results

Metric	Validation	Test
Accuracy	0.8789	0.8897
Precision (weighted)	0.8785	0.8895
Recall (weighted)	0.8789	0.8897
F1-score (weighted)	0.8784	0.8893

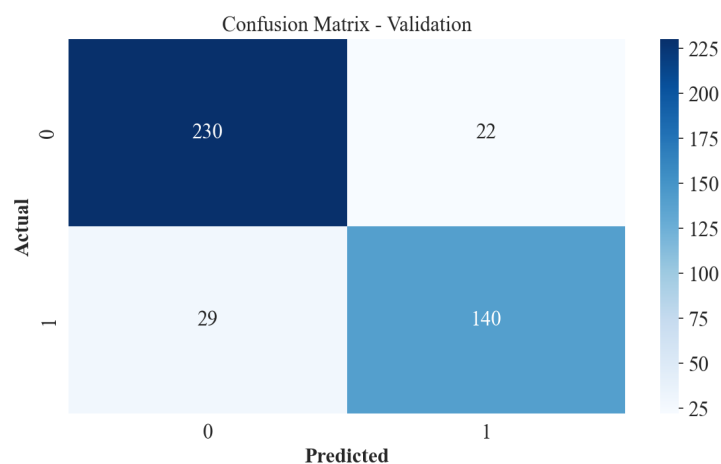


Figure 16: Confusion Matrix – Validation

Inference: The model achieves strong balanced performance, with slightly lower recall on the spam class than on the non-spam class, indicating a modest bias toward predicting "not spam".

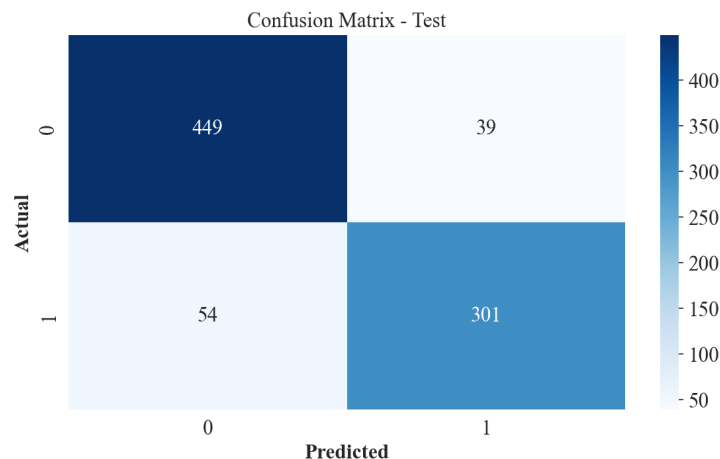


Figure 17: Confusion Matrix – Test

Inference: Test accuracy (88.97%) is consistent with validation, showing the model generalizes well; most misclassifications are false negatives (spam predicted as ham).

10 Discussion

Across all four datasets, Random Forest with Chi-Square/ANOVA-based feature selection provided strong baseline performance, achieving perfect separation on the well-structured Iris dataset and above-0.88 accuracy on the larger Spambase dataset. Performance was weakest on the Diabetes dataset due to class imbalance and physiologically implausible zero values acting as noise. The Loan Amount regression task achieved a high R^2 (0.89-0.91) despite right-skewed and outlier-heavy features, showing tree ensembles are relatively robust to such distributions.

11 Conclusion

This experiment demonstrated the complete ML workflow – library exploration, dataset loading, EDA, preprocessing, feature selection, and model evaluation – using NumPy, Pandas, SciPy, Scikit-learn and Matplotlib across classification (Iris, Diabetes, Spam) and regression (Loan Amount) tasks, correctly identifying the ML task type and an appropriate algorithm for each dataset.

12 References

- [1] Scikit-learn Documentation, <https://scikit-learn.org>.
- [2] Pandas Documentation, <https://pandas.pydata.org>.
- [3] UCI Machine Learning Repository, <https://archive.ics.uci.edu>.
- [4] Kaggle Datasets, <https://www.kaggle.com/datasets>.